Health Deterioration Detection in Hypertensives

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ABSTRACT
Predicting the gradual unraveling of cardiac protection in hypertensive patients is a major challenge for clinicians, who rely on limited data collected sporadically during scheduled patient visits. Sentinel Healthcare has developed a Remote Physiologic Monitoring platform to gather and analyze remotely collected patient data. Patients use provided blood pressure cuffs to take their own vital signs regularly and upload the data to Sentinel's app platform. Healthcare providers can use this data to better provide care and intervention for at-risk patients. In this paper, we report preliminary results on modeling the collected blood pressure and heart rate data to predict if a patient will experience a health deteriorating event in the near future.

Keywords
Time Series Modeling; Heart Rate; Blood Pressure; Remote Patient Monitoring; Hypertension

1. INTRODUCTION
Sentinel Healthcare is a Seattle based business that provides a software platform for comprehensive remote management for hypertension. Sentinel Healthcare is tackling the urgent medical problem of hypertension, particularly in patients with CHF (congestive heart failure). One way CHF and hypertension patients manage their conditions is by vital sign monitoring. We want to detect if vital signs are changing ahead of time and provide signals and indicators to Sentinel Healthcare for effective patient or clinician outreach.

2. BACKGROUND
Blood pressure and heart rate are signs of a heart's function and status. The relationship between cardiac outcome and exercise (walking) is well-established, as is the role of beta blocker medication, to protect the heart. Changes in patient outcome or medication status are coordinated during visits to medical caregivers and often tracked via clinician notes, in text form. Sentinel is endeavoring to combine these data inputs to better provide patient care and valuable analytics to caregivers by:

- Detecting potential risks for patients by vital sign trend
- Extracting meaningful events from clinical notes, such as serious symptom reporting or variation of medicine dosage.

Regarding congestive heart failure (CHF), a major shift in thinking has occurred. In essence, heart failure is a pump failure. Years ago, most doctors thought heart rate-slowing beta blockers would be detrimental to heart failure. However, beta blockers have been shown to enable better, more useful pumping in heart failure patients. This spurred the term "cardio protection." Cardio protection from Beta Blockers might become less effective or ineffective over time, and could be detected by statistical analysis of Sentinel datasets.

3. PROBLEM STATEMENT
The main question our project addresses is: can heart rate and blood pressure be used as a prognostic of a clinical outcome that maps to information captured by doctors' notes on patient treatment and status?

Sentinel Healthcare has provided the UW capstone team with datasets that contain historical data of patients on beta-blockers, with which to produce predictive models, NLP classifiers, and meaningful data visualization for their operational dashboards used by their patient care team. The vitals dataset includes systolic and diastolic blood pressure (SBP and DBP) values and heart rate for each patient, and the Beta Blocker dosage dataset has the type and dosage of Beta Blockers, and date administered for each patient. The clinical notes data identifies specific outcomes for each patient, as tracked by text entered as unstructured clinical notes by doctors or other clinicians.

4. DATA SUMMARY
Sentinel Healthcare has provided datasets that contain historical data of patients on beta-blockers, vitals tracking recorded from their patients, and clinician notes about patients at doctors visits or similar events.

4.1. Vitals data
The vitals dataset includes historical heart rate and blood pressure values for each patient, measured by electronic devices such as MOCACuff. It has 141932 rows and 33 columns corresponding to 1166 patients. After cleaning, the used dataset has the following schema:

Table 1. Vitals data schema

Column	Description	Type	Example
patient_uuid	Patient key	Integer	9988948
time stamp	Medical event date and time	String	2019-11-02 23:59:59.520000

HR	Heart rate	Float	78.0
SBP	Systolic blood pressure	Float	127.0

The vitals dataset was cleaned by removing invalid dates and patients with less than 30 vital recordings, ending up with 49754 data points for 829 patients. After filtering it by selecting rows corresponding to patients under beta blockers, the Vitals dataset contained 48913 data points for 262 patients.

4.2. Notes data

The unstructured notes dataset contains the notes written by Sentinel Healthcare clinicians/staff for each patient. The dataset originally contained 21,370 rows and 7 columns corresponding to 1536 patients, and follows the following schema:

Table 2. Notes data schema

Column	**Description**	**Type**	**Example**
patient_ uuid	Patient key	Integer	9988948
time stamp	Medical event date and time	String	2019-11-02 23:59:59.520000
note	Note written by clinician	String	{'type': 'internal', 'content': 'CFC week 28'}

The notes dataset will be used to validate the heart rate and blood pressure recordings with symptoms and clinical outcomes in order to provide context to the behavior and irregularities of the data. Additionally, it will help avoid misinterpretations or equivocal assumptions when analyzing the data and building a predictive model.

5. APPROACH

5.1. Data Interpolation and Aggregation

Since the data provided on vital sign readings requires patients to routinely submit recordings, there are time-periods where patients have not recorded data or where there are multiple recordings in a given day. Excess data is dealt with by averaging all readings across a day to a single value. Missing data is dealt with by first determining if the missing gap is longer than 15 days. If the gap is longer than 15 days we do not interpolate the values and segment the patient's data into distinct chunks. If the gap is shorter than 15 days we apply an interpolation method called cosinor which is based on the circadian rhythm of the person [3]. This way we generated an augmented dataset for each patient with real data where available, and interpolated data where there are gaps. See figure 1, 2, 3 for an illustration of how missing data is interpolated.

5.2. Models

Once the data has been aggregated, segmented, and interpolated using the methods described above, we apply a number of time series statistical models trained on each patient's individual data to forecast their vital signs. Below we will describe each of these models in more detail.

5.2.1 Long Short Term Memory (LSTM)

A Long Short-Term Memory network or LSTM network is a type of recurrent neural network used in deep learning. They are trained using Backpropagation Through Time and they overcome the vanishing gradient problem. Unlike RNN neurons, LSTM uses memory blocks connected through layers. LSTM models are well suited to both univariate and multivariate time series data [2].

LSTMs are sensitive to the scale of the input data, so it is often good practice to rescale or normalize the data to the range of 0-to-1. This project uses the MinMaxScaler preprocessing class from scikit-learn to do so.

The LSTM network expects the input data (X) to be provided with a specific array structure in the form of: [samples, time steps, features], so this model transforms input train and test data to this expected structure. This network has a visible layer with 1 input, a hidden layer with 100 LSTM blocks, and an output layer that makes n_steps_out prediction, which is set to 7 to represent one week of predicted vital signs. The network is trained for 50 epochs, which has a significant impact on training time; for our cleaned dataset of 883 patients, this model takes between six and eight hours to run locally.

Once the model is fit, performance of the model is evaluated on the train and test datasets. and predictions are inverted before calculating error scores to ensure that performance is reported in the same units as the original data, heart rate bpm (beats-per-minute) or blood pressure mmHg (millimeters of mercury).

5.2.2 Vector Autoregression (VAR)

Vector Autoregression is a multivariate forecasting model used when two or more time series influence each other, heart rate and systolic blood pressure time series in this case. The model cannot handle analyzing the time series data of multiple patients at the same time, therefore it is deployed on a patient level, and the resulting performance metrics are aggregated in order to evaluate the overall performance of the model.

There are several tests that are necessary to validate the usage of the model on the patient's time series data, namely, the Granger's Causality Test, Johanssen's Cointegration Test, the ADF Test (Augmented Dickey-Fuller Test), and the Durbin Watson Statistic.

The Granger's Causality Test is used to test the relationship between heart rate and blood pressure to see if the past values of one feature influences the values of the other one. The null hypothesis of the test is that the coefficients of past values in the regression equation is zero, therefore if the p-value for the test is less than 0.05, the null hypothesis is rejected, meaning that there is interaction between the two variables. 61.03% of the patients exhibited patterns indicating that the past values of systolic blood pressure affect the values of heart rate, and 61.85% of the patients showed that the opposite relationship is also true, where the past values of systolic blood pressure affect the values of heart rate.

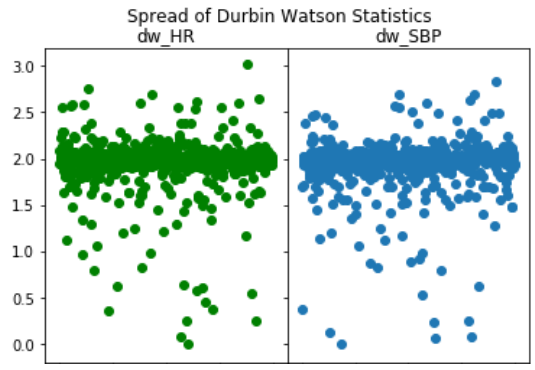
Johansen's Cointegration Test is used for statistically significant relationship between two or more time series. Around 46% of the patients showed between variables, which might be explained by the non-stationarity of the data of some patients. [4]

The Augmented Dickey-Fuller Test is used to check the stationarity of the time series. If the time series is not stationary, the dataframe is differentiated and the test is run again. This process is repeated until all the variables are stationary. 56% of the patients show seasonality in their data, while 40% required differentiating the data once to achieve seasonality in both heart rate and systolic blood pressure.

The order (P) of the VAR model is the number of differentiations required in order to convert a non-stationary time series into a stationary one, and is obtained by iteratively fitting increasing orders of the model, and choosing the one with the lowest Akaike information criterion (AIC). The model is fitted by the selected order and used for forecasting.

The Durbin-Walter Statistic calculates the serial correlation of the residuals of the fitted model. It ranges from 0 to 4, and if it's close to 2, it means there is no significant serial correlation, meaning that there it rejects the null hypothesis that there is some pattern that is still left to be explained by the model. According to Figure 4, most patients present scores close to 2, more specifically, around 88% of the patients show scores between 1.75 and 2.25.

Figure 1: The plot shows the output for the Durbin Watson Statistic for each patient. The x values are randomized to spread the scatter points along the x-axis.



After obtaining the forecasted values, the differentiation of time series of the patients that had non-stationary variables is inverted in order to show the real forecast that is used to evaluate the performance of the model against the test set.

5.2.3 Seasonal Autoregressive Integrated Moving Average (SARIMA)

SARIMA is an extension to the popular ARIMA (Auto Regressive Integrated Moving Average) technique used for most time series analysis. The drawback to ARIMA is that it does not support time series that have seasonal components. Seasonal ARIMA, or SARIMA, supports time series that have both trend and seasonality.

In brief, ARIMA models contain hyper parameters for Auto Regression (AR), Differencing (I) and Moving Average (MA). SARIMA adds three additional hyper parameters for these factors for the seasonal component. In addition, it also contains a fourth hyper parameter for the number of time steps for a single period. The hyper parameters, therefor, are:

- p: Trend autoregression order
- d: Trend difference order
- q: Trend moving average order
- P: Trend autoregression order
- D: Trend difference order
- Q: Trend moving average order
- m: The number of steps for a single seasonal period

The model is notated as SARIMA(p,d,q)(P,D,Q)m

For the hyperparameters, we performed a grid search and picked the combination that results in lowest AIC (Akaike Information Criterion). Lower AIC implies high log likelihood that we see the observed data, given the model.

5.3 Detecting Deterioration with Flags

The flag-raising algorithm parses through a patient's time series containing 30 days of continuous data. It raises a flag whenever conditions regarding health deterioration are met. There are 8 different flags, which can be binary values (patient showing increased heart rate) or numerical values (number of times systolic blood pressure surpassed 180mmHg).

- rising_HR: True if the difference of the mean values of heart rate of the last 10 days is greater than 10. False otherwise.
- lowering_HR : True if the difference of the mean values of heart rate of the last 10 days is lesser than -10. False otherwise.
- rising_SBP: True if the difference of the mean values of systolic blood pressure of the last 10 days is greater than 5. False otherwise.
- high_BP: True if systolic blood pressure readings between 140 and 179 occur over 3 times in the last week.
- min_HR: Number of times heart rate was observed to be lesser than 55.
- max_HR: Number of times heart rate was observed to be greater than 100.

- min_SBP: Number of times systolic blood pressure was observed to be lesser than 90.
- max_SBP: Number of times systolic blood pressure was observed to be greater than 180.

The performance of the flag-raising algorithm on the three models was measured by calculating the accuracy and F-1 scores for the binary flags, and the Mean Absolute Percentage Error (MAPE).

6. RESULTS

Table 3 outlines the final error rates for HR prediction as well as SBP prediction for 3 models using MAPE, MAE, RSME. Generally, LSTM performs best in predicting SBP while VAR performs best in predicting HR. We believe this is because VAR considers the interaction effects between HR and SBP and SBP is harder to predict given HR while the opposite is not true, according to Johansen's cointegration test.

Table 3. Performance indicators for the models.

	LSTM	VAR	SARIMA
HR_mape	0.03	0.04	0.05
HR_mae	4.40	3.22	6.13
HR_rsme	5.47	4.05	7.75
SBP_mape	0.03	0.05	0.05
SBP_mae	2.57	5.96	3.56
SBP_rsme	3.21	7.48	4.58

Table 4: Performance on Flag-Raising indicators for model

Flags	LSTM	VAR	SARIMA
acc_rising_HR	0.02	0.99	0.02
f1_rising_HR	0.01	0.50	0.01
acc_lowering_HR	0.98	0.99	0.98
f1_lowering_HR	0.00	0.67	0.00
acc_rising_SBP	0.89	0.91	0.89
f1_rising_SBP	0.02	0.57	0.00
acc_high_BP	0.77	0.93	0.78
f1_high_BP	0.00	0.84	0.00
mae_min_HR	0.27	0.15	0.26
mae_max_HR	6.81	0.10	6.71
mae_min_SBP	6.50	0.06	6.38
mae_max_SBP	0.02	0.04	0.03

Table 4 presents the performance metrics of the three models using the flag-raising algorithm as masking for the predicted values. Generally, VAR performs best in finding health deterioration signals. The table shows that the LSTM and SARIMA models have difficulty in finding true positive values, which explains why the F-1 scores are low.

Figure 2: Histogram of errors in SBP by model.

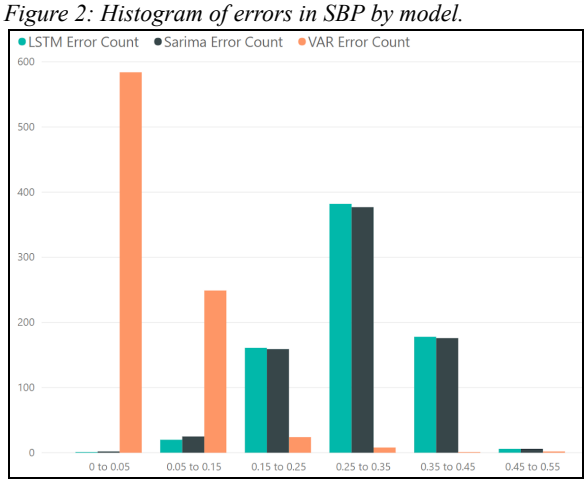


Figure 3: Histogram of errors in HR by model.

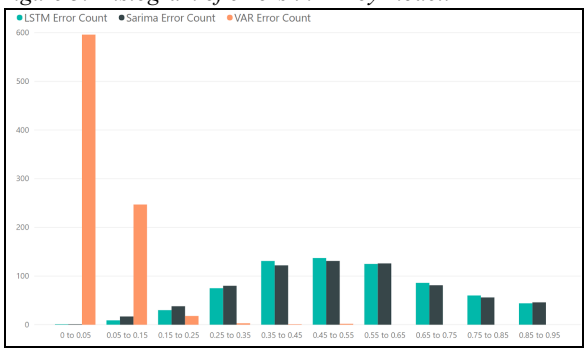


Figure 3 and Figure 4 show the histograms of the error rate of the models for HR and SBP recordings for the last 10 days of the patients' time series respectively. LSTM and SARIMA show a normal distribution in both cases, while VAR shows a logarithmic distribution.

7. CONCLUSION

Our models show potential for time series modeling on vitals data. Initial results were promising, and we think these models are a framework to which additional signals such as Beta Blocker dosages can be added for robustness.

Cleaning of data and interpolation of missing values demanded significant effort which could be mitigated as Sentinel's platform matures and provides additional data, including more densely populated vitals data and related information, such as ambulatory data.

The UW capstone team forsees the following next steps:

- Incorporate beta blockers data with half-life of each medicine for a better understanding of deterioration of health in relation to patient medication [5].
- Refine data visualization to be operationalized by Sentinel Healthcare platform, including multiple

readings per day per patient in a line chart mixed with a scatter showing vital sign means.

8. ACKNOWLEDGEMENTS

We would like to thank our sponsors and guides at Sentinel Healthcare: Nirav Shah, Noah Manders and Naman Sharma. We would also acknowledge the guidance of our capstone project professor, Megan Hazen.

9. REFERENCES

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10. APPENDIX

Figure 2: Example of Prediction and Performance Metric

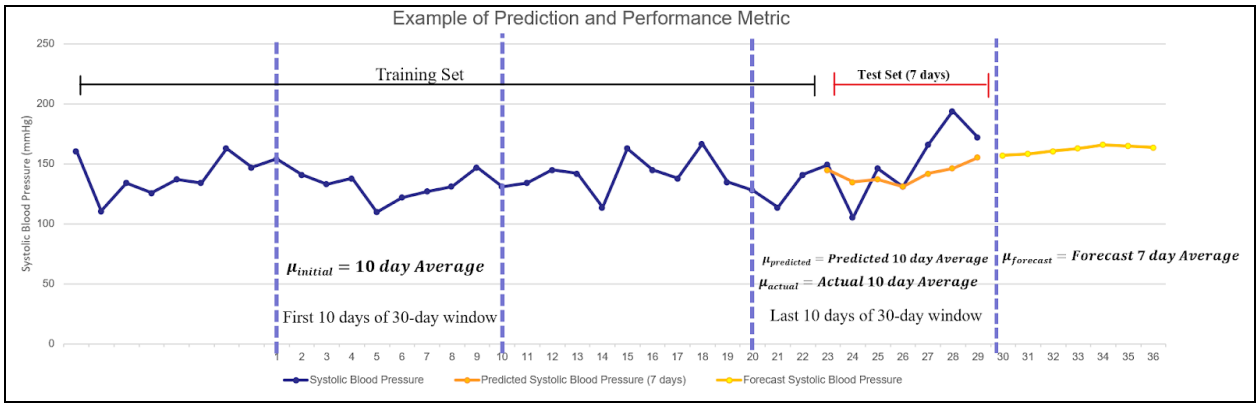


Figure 3: Example of Cosinor Model

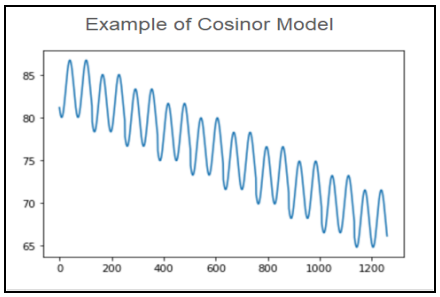


Figure 4: Example of Data Interpolation

